



SECP1513-03 TECHNOLOGY AND INFORMATION SYSTEM

Design Thinking Project Report

Product Name: PlayMatch – A Smart Booking and Athlete Partner Matching System

Group Members: 5 Students

Lecturer: TS. DR. MUHAMMAD IQBAL TARIQ BIN IDRIS

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1.0 Introduction

What motivates us to do this project is the problem, as our team realized it, is the "Lonely Athlete" and the "Booking Maze." Someone may love being active, but if they don't have any friends who love being active in the same way, then they're left out since nobody is around to play with them.

We also found that in today's sports world, it's difficult to join in because we lack instructions. Booking a court involves many phone calls, texts, and unspoken rules. This can be very difficult. Technology should be used to make it simpler, but it shouldn't be used to complicate things. PlayMatch is intended to be a tool for these "solo players" to join easily by providing a platform that is easy to use to book a court with new athletic friends.

i) -Empathies	We start by putting ourselves in the shoes of athletes and local sports club organizers to uncover "hidden" pain points. By carrying out user interviews and field observations, we can really understand what the barriers to access training means to them and what their social needs are.
ii) -Define	In this phase, we synthesize research to determine what exact digital gaps exist, such as a lack of personalized performance metrics or inclusive solutions for under-served audiences. Here, we shift from making general observations to creating a defined problem statement centered around democratizing and enhancing the sporting life for all individuals impacted.
iii) -Ideate	In this phase, the emphasis lies in the "blue-sky thinking" process. We brainstorm numerous technological aspects, ranging from the performance analysis capabilities of AI to the engaging fan experience tools. Using the process of mind mapping, we sift through the ideas to identify the most feasible technological solutions for community linkages and wellness.
iv) -Prototype	For this phase, we concentrate on the visual aspects of our prototype. We build on the concepts we have developed to create a complete interface flow from the login page to the output of data. Basically, this is a non-technical process where we focus on the usability of the interface without investing in coding.
v) -Testing	The final stage of the project is to place our interface design in front of real-life sports fans. Through this process of testing how users are able to interact with the design modules, we are able to collect feedback that will perfect the design of the digital tool and empower the fans.

2.0 Detail Steps

In December 2025, we were tasked with prototyping PlayMatch - Smart Booking & Athlete Partner Matching System. We established that there could be a client for PlayMatch - a client would be a student who would like to book the court without going to the sports hall and also find playing partners. Here, we need two models: one that accurately checks the court availability for booking ahead, and one that matches students looking for partners.

2.1 Empathies

To initiate PlayMatch, we identified some of the root frustrations in the local sporting environment. We carried out a discovery interview with James Sim Chia Tai, an UTEAM athlete, to identify some of the key underlying aspects of setting up matches between amateur participants. We also investigated the hindrances of finding a compatible match as well as a venue to do so.

2.2 Define

Second, we synthesized the information obtained from the empathetic processes above. We discovered that there were two problem areas that the platform could address:

- (1) There was a large gap between the level of match that the player could undertake;
- (2) There was a problematic booking system that involved multiple calls around the venues for court availability.

Our mission was to take these two areas of difficulty and turn them into an optimized digital solution.

2.3 Ideate

Zeroing in on the problem of matching disparity and booking congestion, we started brainstorming innovative ideas ranging from AI-powered skill matching to real-time GPS-based facility location tracking. We sought to design these products to make booking easier and to develop an active sports community. The central ideas of Athlete Matchmaker and Instant Book Dashboard were formed to give a structure to PlayMatch.

2.4 Prototype

During prototyping, we converted our concepts into a visual architecture and a flow of a user journey. Each of our team members is responsible for modules such as setting up athlete profiles, searching partners, and safe checkout of facilities. Our prototype displays the overall look and feel of PlayMatch, from matching to booking a court.

2.5 Testing

The PlayMatch interface was also tested by simulating fundamental tasks like filtering for a badminton partner or creating a digital receipt for booking. The issue of friction when navigating through the interface was assessed, along with feedback received regarding its understandability. The final design will perfectly accompany the sports community.

3.1 Problem: The Offline Barrier in Modern Sports

In contemporary sports, playing organizational games is friction because of:

- Unstructured processes
- Various WhatsApp groups
- Venue repeats call for slots Lack of real time synchronization results in:
- Skill/Schedule/Location Gap: Lack of partners for games, resulting in uncompetitive games or last-minute changes.
- Booking Blind Spot: No visibility of venue availability. Players can come or ask the booker to check the ledger.

3.2 Solution: PlayMatch - The Unified Sports Ecosystem

PlayMatch is an intelligent online system that brings together players and facilities, acting as an intermediary. The main features include:

- Athlete Matchmaker: Profiles that match players in the area based on skill levels, sport type, and location.
- Instant Sync Booking: Real-time court availability on a dashboard with secure booking, receipts, and reminders for proper usage of popular locations.

Automating administrative tasks, PlayMatch allows flexible access to a dynamic sports culture.

3.3 Team Working: Synergy Through Design Thinking

The team is made up of five members. We applied the principles of the Design Thinking approach as informed by user evidence.

- Discovery and Research: Problem identified through interviews conducted on James Sim Chia Tai. - Role Assignment: duties based on strengths regarding UI/UX and research.
- Continuous Synchronization: This allowed regular sprints to show design problems early and kept PlayMatch integrity intact.

This kind of communication facilitates timely digital prototyping.

4.0 Design Thinking Assessment Points

Assessment is an important concept in the design thinking cycle in order to remain on track regarding the solution and the needs of the users. PlayMatch – A Smart Booking and Athlete Partner Matching System in UTM is an attempt to provide the students with a rapid facility to book sports courts and partners according to skill, timing, and preferences.

Empathy phase: We conducted interviews and surveys with students to gain an understanding of the challenges they face. This was done with the objective of identifying challenges such as the difficulty of booking a venue and partners with the same level of skill.

Define phase: We analyzed and classified user data into precise problem statements in order to define system requirements before moving forward.

Phase of ideation: Ideas were generated by the team to solve a problem, and some of the features identified were intelligent athlete matching based on skill and timing. We also assessed the feasibility of each selected project to meet user needs.

Prototype stage: This involved the creation of an easy-to-use booking and partner search system that includes sports facility booking, student details, and partner recommendation algorithms, with an emphasis on ease of booking as well as appropriate partner matching.

Test Phase: Students tested our prototype, and we gathered input on usability, accuracy of matches, and functionality. Improvements in user interface design and optimization of the matching algorithm have been incorporated so that our final project meets requirements.

5.0 Design Thinking Evidence

5.1 Empathy Phase

During Empathy Phase, we engaged directly with users to understand their needs. We conducted in-depth interviews with UTM students, James Sim Chia Tai, a state player of Ping Pong from Sarawak. Besides that, we have also distributed a Google Form survey to gather quantitative feedback. From these interactions, we were able to identify the core challenges students face with the current sport hall booking process and the difficulty of finding teammates.



Figure 1: Interview with James Sim Chia Tai

1. How often do you currently play sports?
2. On a scale of 1 to 5, how difficult is it to find an available court/venue during peak hours?
3. On a scale of 1 to 5, how difficult is it to find partners or opponents at your skill level?
4. What is your biggest frustration when organizing a game?
5. How do you currently book your sports venues?
6. How likely would you be to use this feature to play with strangers?
7. Which of the following features is MOST important to you?
8. If PlayMatch offered a "Reliability Score" to ban users who ghost/cancel last minute, would that make you trust the system more?

Figure 2: List of questions during the interview

5.2 Define Phase

During Define Phase, we reviewed all the information collected from the interview sessions and Google Form. We gained a better understanding of the problems students face, particularly the hassle of manual booking and the absence of a partner-matching system. This insight helped us design our solution to better meet their specific needs and challenges. Below are the results from the Google Form.

Which sports do you play most frequently?
(33 条回复)

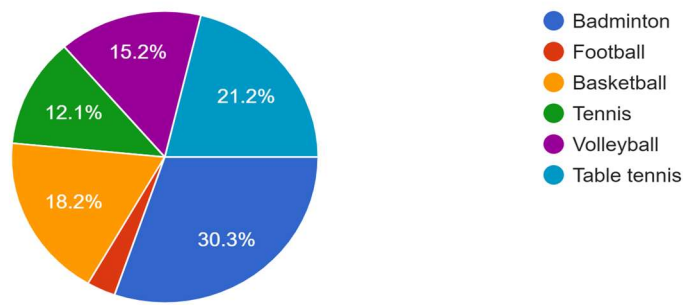


Figure 3.1

How often do you currently play sports?
(33 条回复)

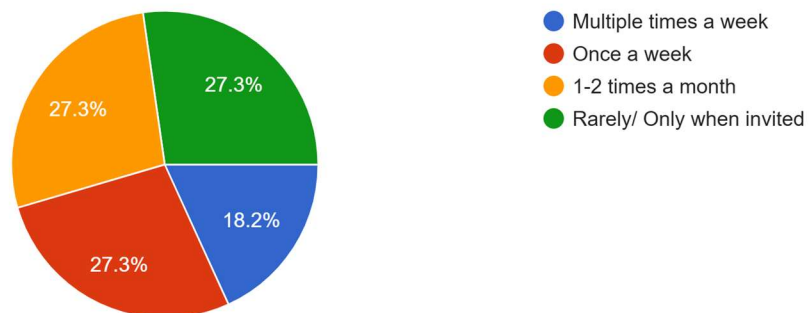


Figure 3.2

How difficult is it to find an available court/ venue during peak hours?

(33 条回复)

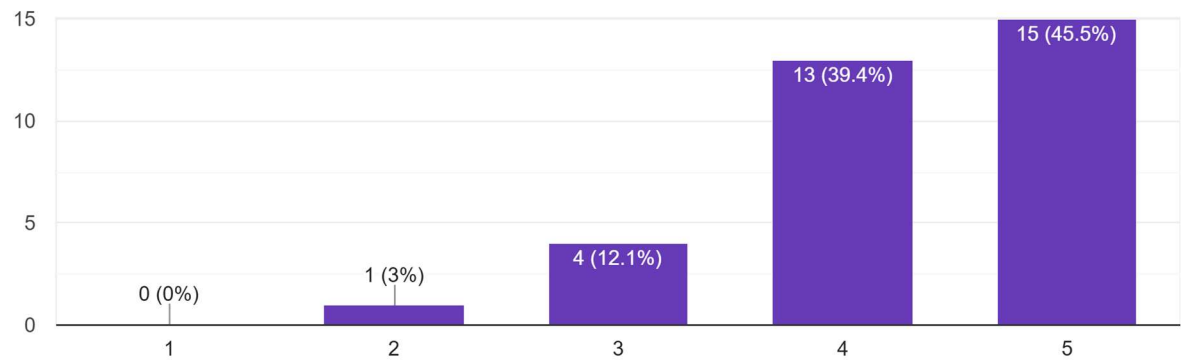


Figure 3.3

How difficult is it to find partners or opponents at your skill level?

(33 条回复)

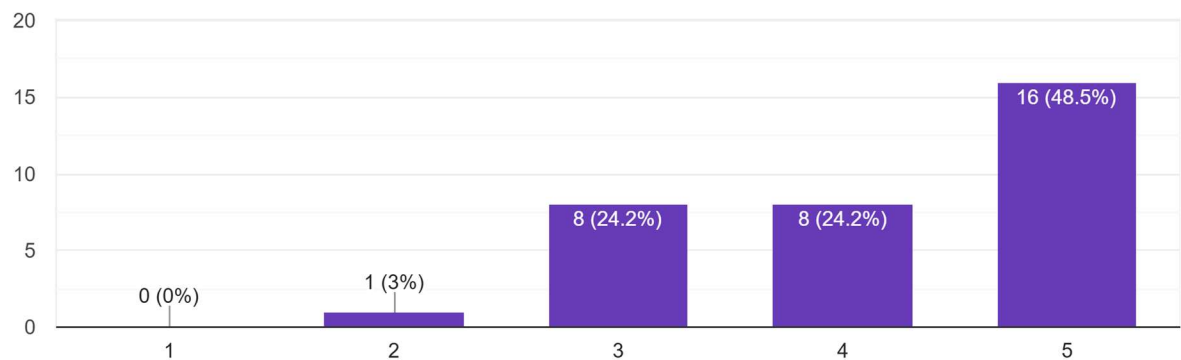


Figure 3.4

What is your biggest frustration when organising a game?

(33 条回复)



Figure 3.5

PlayMatch allows you to book venues instantly. How do you currently book your sport venues?

(33 条回复)

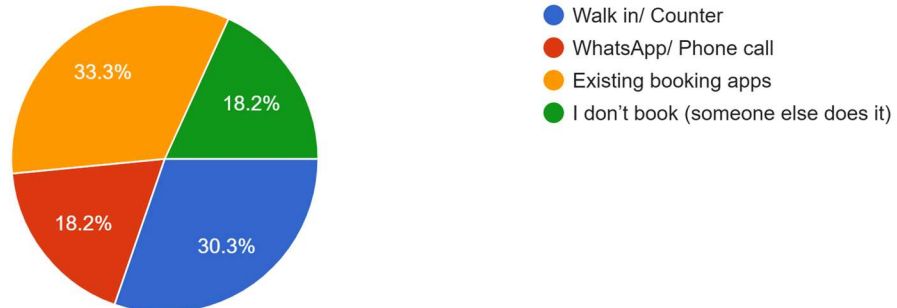


Figure 3.6

PlayMatch includes a “Smart Matching” feature to pair you with players of similar skill. How likely would you be to use this feature to play with strangers?

(33 条回复)

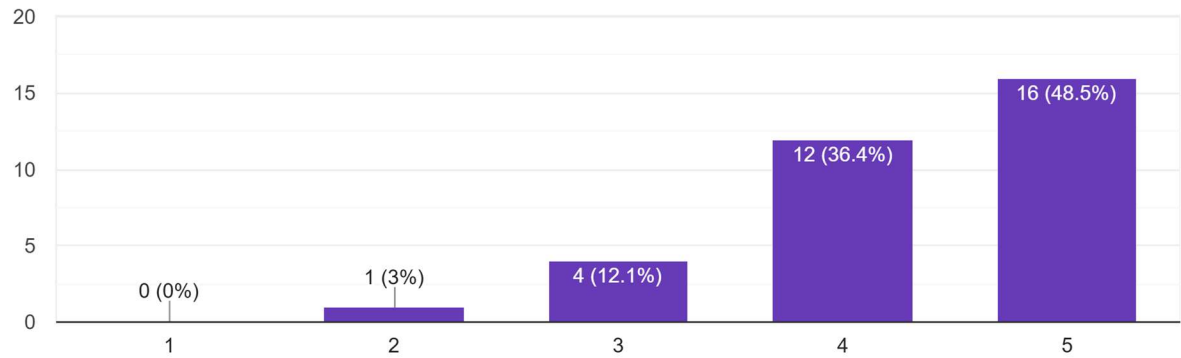


Figure 3.7

Which of the following festure is MOST important to you?

(33 条回复)

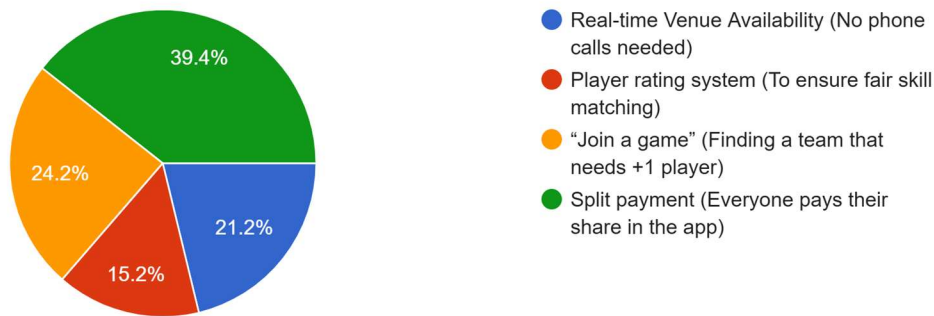


Figure 3.8

If PlayMatch offered a “Reliability Score” to ban users who ghost/ cancel last minute, would that make you trust the system more?

(33 条回复)

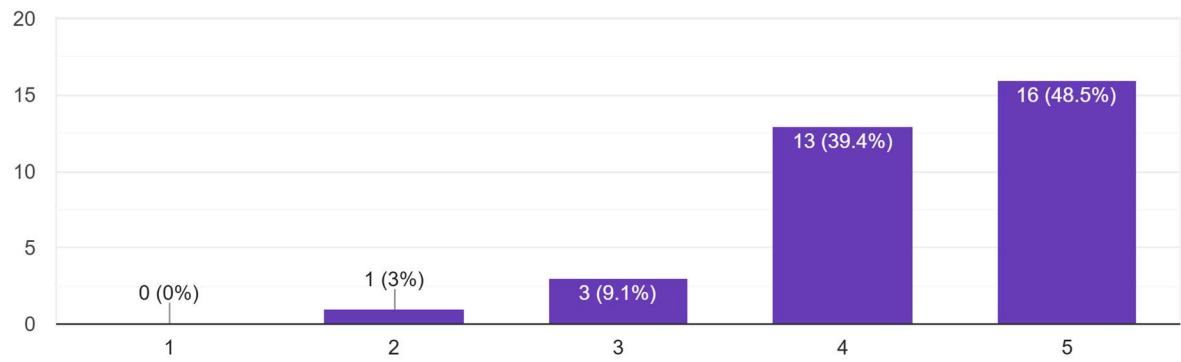


Figure 3.9

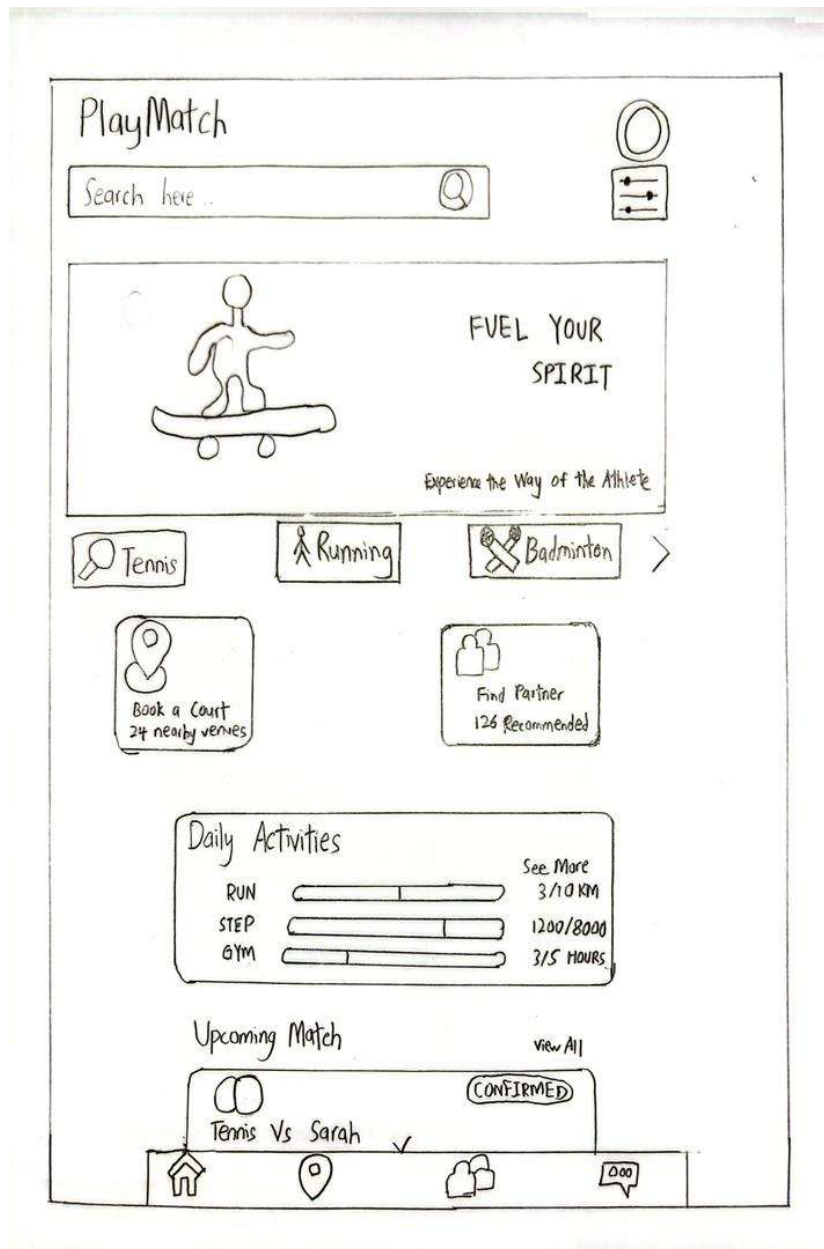
5.3 Ideate Phase

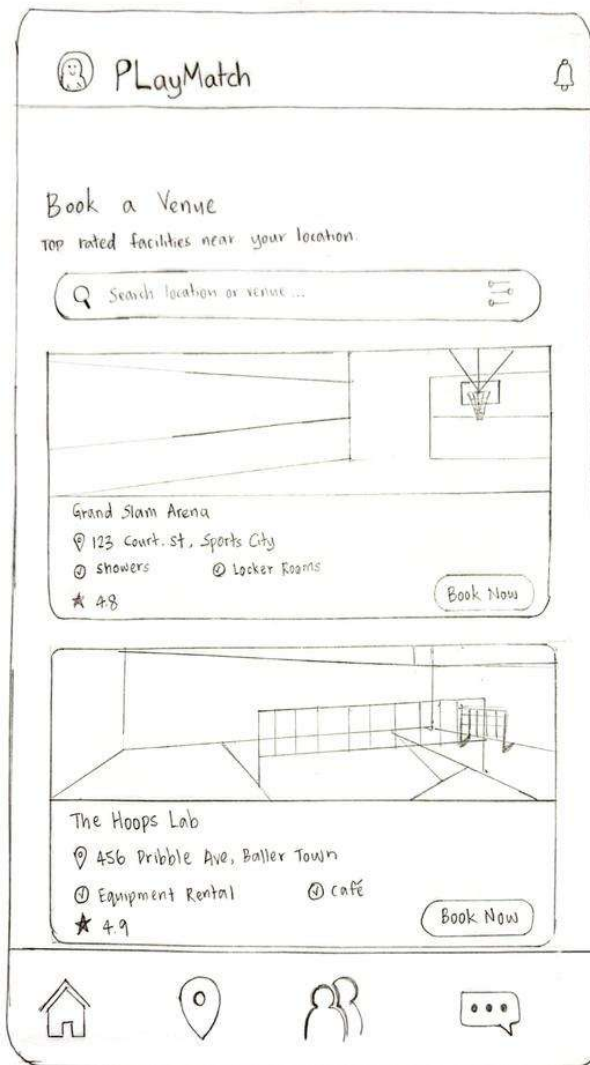
In this phase, we collaborated as a team to find the best solution for the problems we identified. We brainstormed various possibilities and ultimately decided to implement a smart system that displays court availability for booking and includes a feature to match players with partners. We found out that current **Playmatch** app by Nodirbek Razzokov is primarily a football-focused tool for team balancing and group coordination, mostly popular in Uzbekistan. In contrast, ours **PlayMatch** is a comprehensive sports ecosystem management tool.

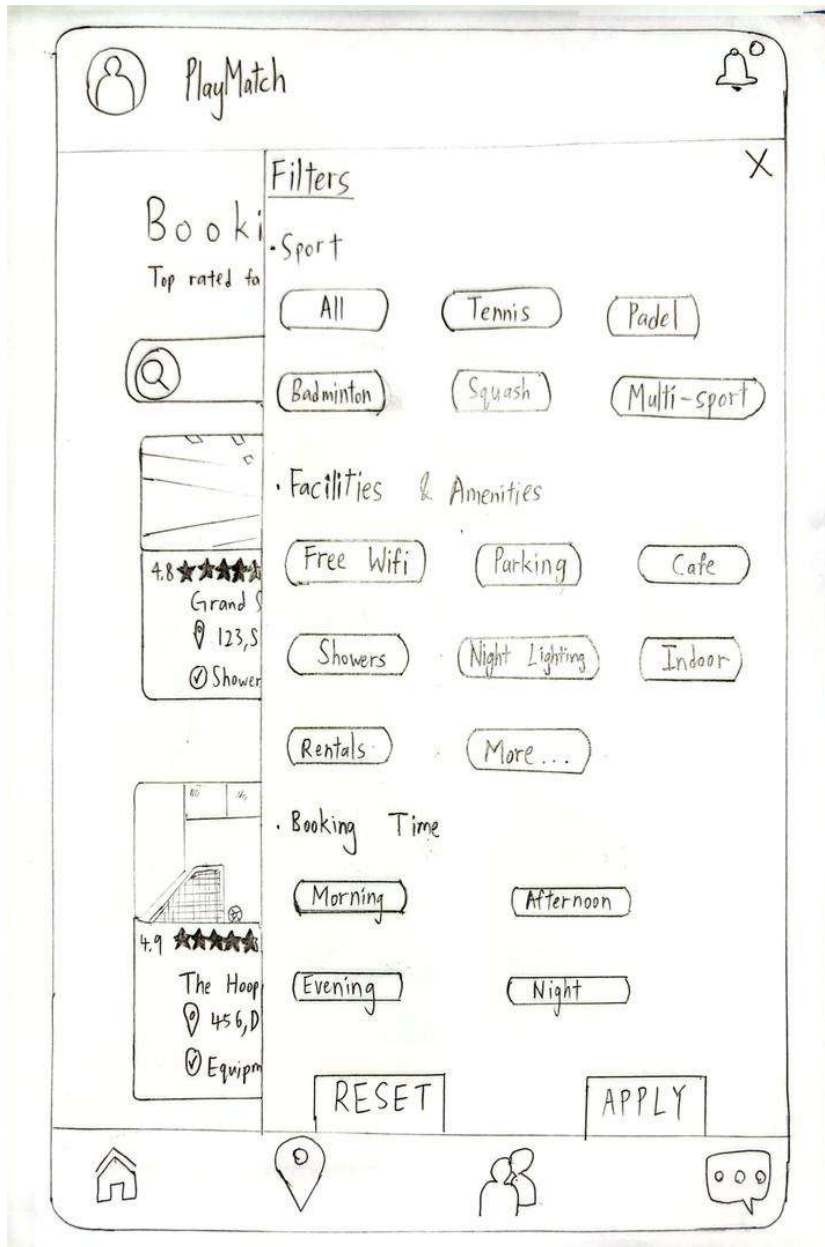
Feature	Playo (App Store - playo)	PlayMatch (Smart Booking & Matching)
Primary Goal	Focus on games finding and players matching.	A smart venue booking and partner-finding ecosystem.
Partner Matching	Certified coaches discovered.	Skill-based matchmaking to find new partners outside your social circle.
Venue Booking	Venue discovering based on location.	Direct facility booking with real-time availability and digital receipts.
Penalty System	None mentioned; relies on social pressure in groups.	Strict Penalty System: Last-minute cancellations lead to money and "trust point" deductions.
Trust/Credibility	Basic player ratings based on post-game votes.	Trust Score: A formal metric impacted by reliability and attendance.
Core Workflow	Telegram-integrated attendance and team creation.	Complete interface flow from Login → Matching → Booking → Output .

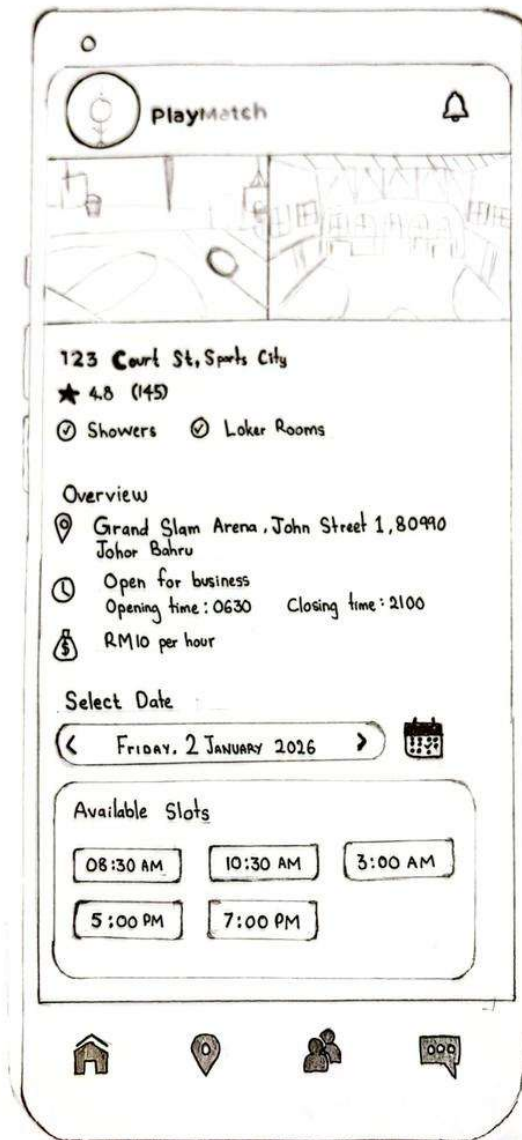
5.4 Prototype Phase

In the Prototype phase, we utilised the ideas and feedback collected from students as a guideline for our design. We transformed these insights into a functional prototype, ensuring that features like court booking and player matching directly addressed the users' specific needs.









PlayMatch

Finding a Match?

Connect with athletes who match your skill level and availability

Sport

Tennis

Level

Intermediate

What are you looking for?

e.g. Someone to practice rallies with on Saturday morning. Chill vibes but competitive

POST YOURSELF

Suggested Partners

Alex

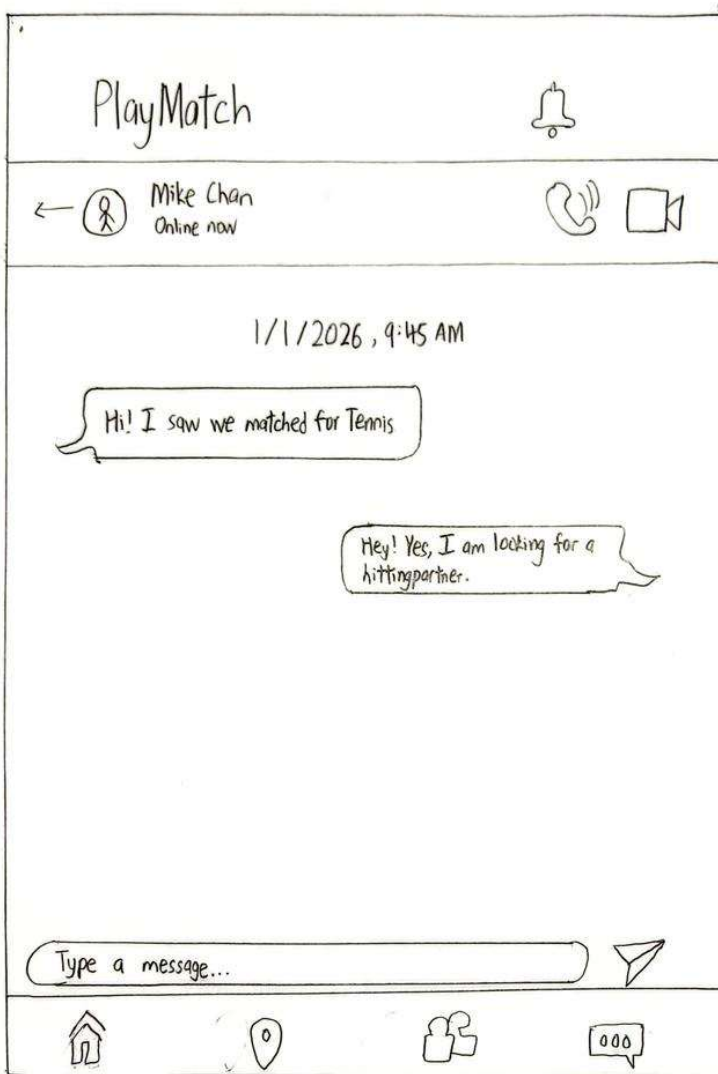
"Looking for high-intensity tennis rallies."

TENNIS

PADEL

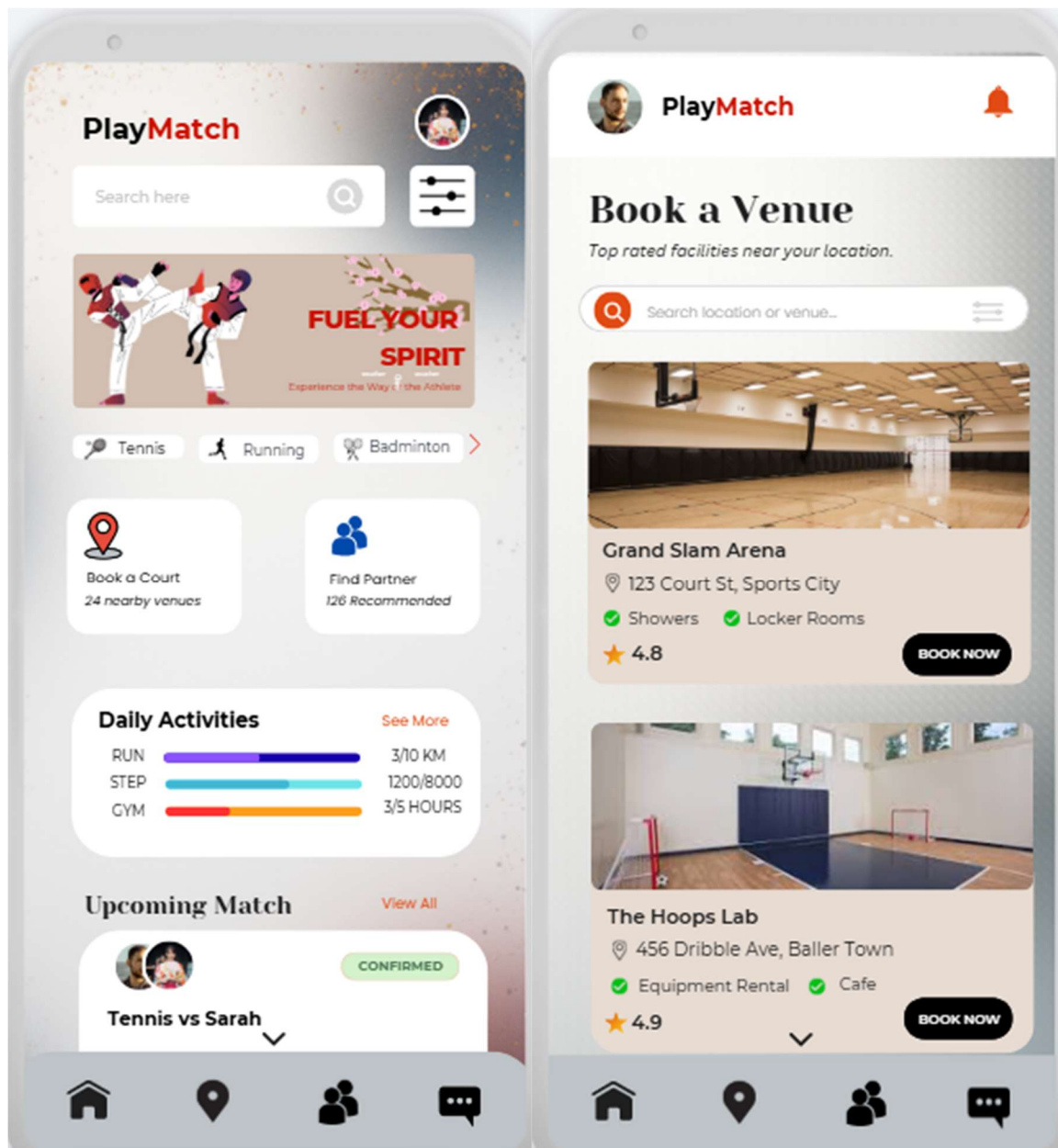
ADVANCED

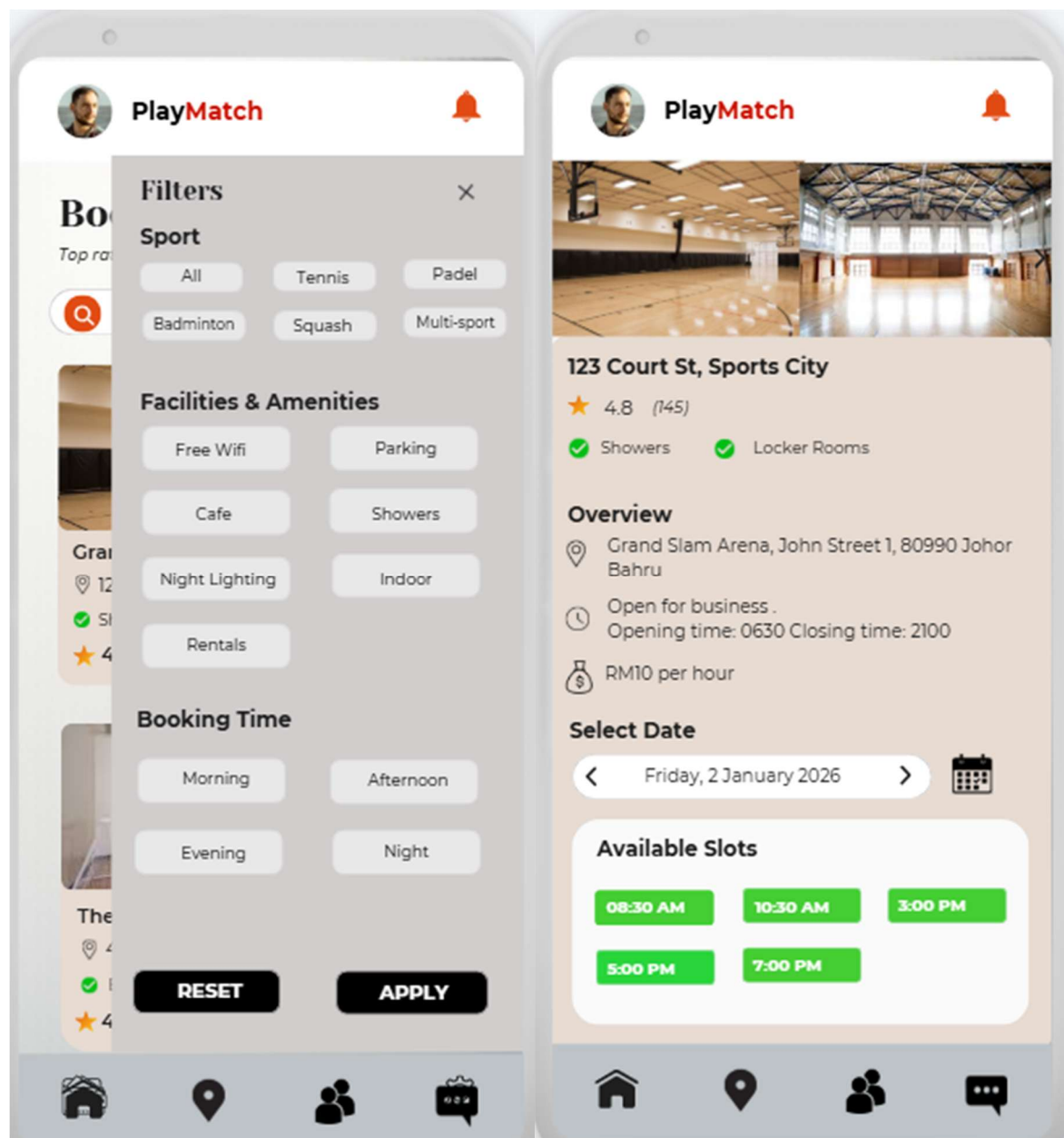
CHAT

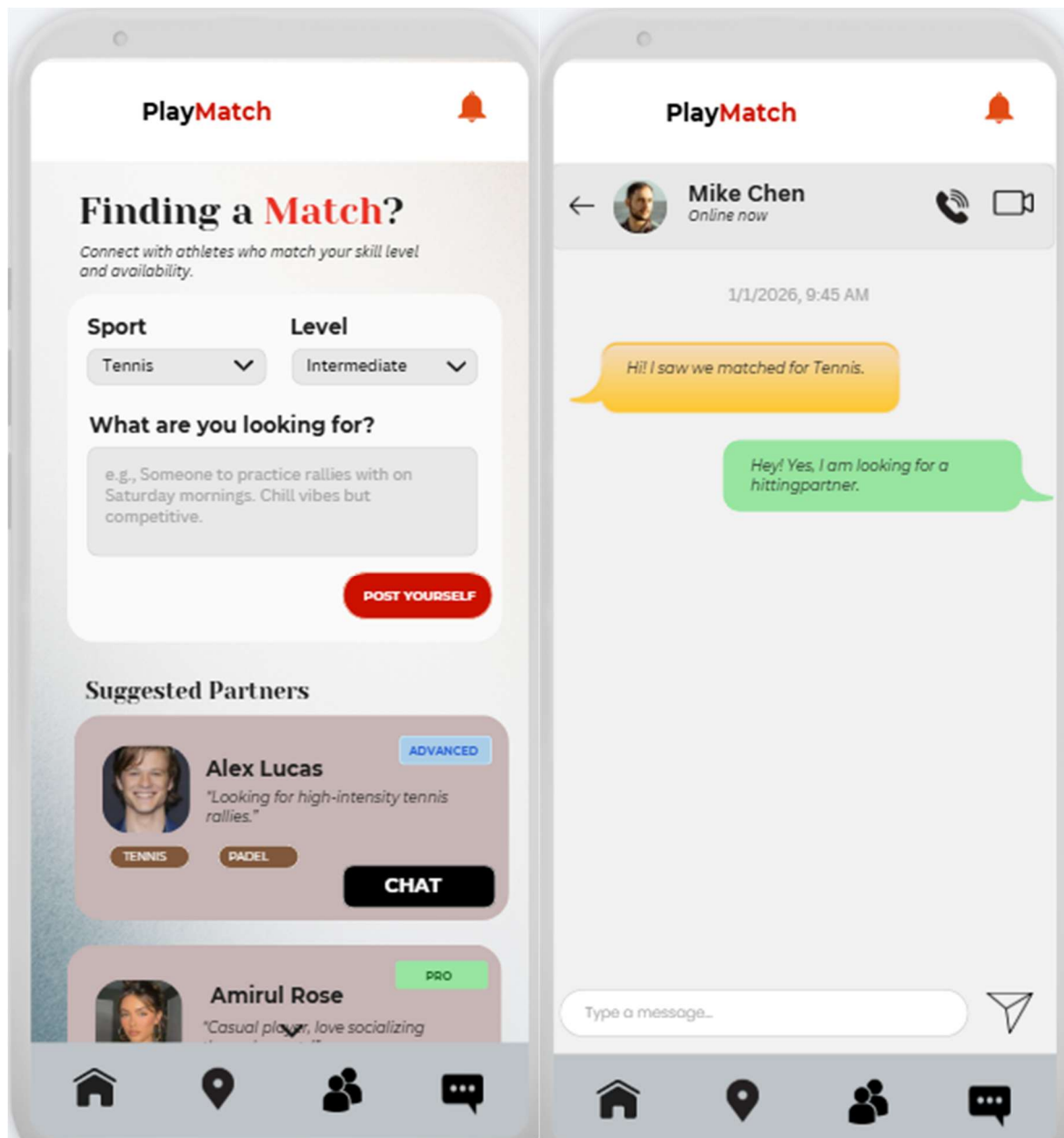


5.5 Prototype Testing Phase

In the Testing Phase, we asked several users to test our prototype. Most of them provided positive feedback and were impressed by the system's functionality. One user specifically stated that this would be very useful for **booking sports courts and finding teammates**, ultimately saving them significant time.







6.0 Reflection

1. LEE JIAN LIANG

What is your goal/dream with regard to your course/program?

My dream is to be involved in the field of Computer Science, particularly in developing intelligent and efficient systems. Through my Data Engineering course, I aim to build strong technical skills in data processing, system design, and problem-solving, which will support my long-term goal in the computer science field.

How does this design thinking impact on your goal/dream with regard to your program?

This design thinking project helped me understand the importance of user-centered system development. Through this project, I have learned how to identify user needs, analyze problems, and design suitable solutions. The process improved my problem-solving, teamwork, and communication skills, which are essential for my future career in the software industry.

What is the action/improvement/plan necessary for you to improve your potential in the industry?

To improve my potential in this industry, I will strengthen my programming skills and gain more hands-on experience through projects. I will also improve my social and communication skills for a better future career.

2. TEOH WEI CHENG

What is your goal/dream with regard to your course/program?

My goal is to become a professional and expert in Data Engineering, which can let me to earn high salary in future.

How does this design thinking impact on your goal/dream with regard to your program?

This design thinking project helped me to track on real-world problems and find the ways to solve them. I learned to communicate with my team members and find the best solution for the design thinking.

What is the action/improvement/plan necessary for you to improve your potential in the industry?

To improve my potential in this industry, I will strengthen my programming skills and try to take part in real-world project. It enhance my critical thinking and let me stand out on the stage.

3. THURGA DEVI A/P PONNAMBALAM

What is your goal/dream with regard to your course/program?

My objective is to become a well-rounded professional in the area of Computer Science, with a focus on creating data-driven systems that are dependable. As such, I am hoping to establish a solid foundation through studying Data Engineering for handling large amounts of data, for developing efficient systems and for using logical thinking when solving real-world issues. The Data Engineering Course is a significant step in meeting my long-term goal of developing innovative technological solutions.

How does this design thinking impact on your goal/dream with regard to your program?

As a result of using the Design Thinking methodology, I have realized that while systems must be technically correct, they must also provide value to their users. I have gained the ability to view problems from the users' perspectives, to clearly define requirements and to develop solutions to meet users' needs. I have developed greater creative capabilities; collaboration skills; and analytical capabilities through completing the PlayMatch Project and this is essential for developing a career in Computer Science and Software Development.

What is the action/improvement/plan necessary for you to improve your potential in the industry?

To further improve my readiness to enter the industry, I intend to continue improving my technical knowledge, specifically with respect to programming and tools used to manage data. I plan to increase my opportunities for hands-on experience, by completing additional projects and by participating in collaborative group activities. Furthermore, I plan to improve my communication and interpersonal skills. These are essential to facilitate effective collaboration within a professional work environment.

4. WONG QI SHEUN

What is your goal/dream with regard to your course/program?

To be honest, I initially thought this course would focus mostly on data collection and processing. I didn't expect to dive so deep into Computer Science, which makes the next four years feel like quite a challenge. But I'm going to give it my best and try to find the fun in learning these skills. My goal is to graduate feeling like I've truly mastered something new and to add a meaningful achievement to my life's journey.

How does this design thinking impact on your goal/dream with regard to your program?

It has taught me how to actually spot problems and come up with real solutions. It really sharpened my critical thinking and teamwork skills. I'm hoping that these are the kinds of abilities that will give me a big head start in the future.

What is the action/improvement/plan necessary for you to improve your potential in the industry?

I need to keep pushing myself and stay persistent, especially in my weaker subjects, so I can really find my footing in this industry later. Beyond just the technical side, I also hope to build a genuine passion and interest in this field as I go along.

5. WONG ZI PING

What is your goal/dream with regard to your course/program?

I aspire to be employed in the sphere of Computer Science, especially the creation of intelligent and efficient systems. With my course in Data Engineering, I will develop good technical knowledge in data processing, system design, and problem-solving, which will help me in my long-term aspiration in the computer science sphere.

How does this design thinking impact on your goal/dream with regard to your program?

The design thinking project made me realize that the system development process should be user-centered. The PlayMatch project helped me to understand how to determine the needs of users, analyze, and design appropriate solutions. The process enhanced my communication,

problems solving, and team-work skills, which are critical in my future profession in the data industry.

What is the action/improvement/plan necessary for you to improve your potential in the industry?

I must improve the learning habits in which I am selfdriven and become more active in learning new technologies. I will also develop a better career focus through learning the trends in the industry and my strengths. Moreover, I will work towards becoming more time-sensitive and flexible to cope with the issues in a rapid changing environment.

7.0 Task Distribution

Members	Task
LEE JIAN LIANG A25CS0078	- Report Writing (Detail Steps & Design Thinking Assessment)
TEOH WEI CHENG A25CS0150	- Interview people & Creating Survey form - Design Thinking Evidence
THURGA DEVI A/P PONNAMBALAM A25CS0368	- Video editing - Record video
WONG QI SHEUN A25CS0161	- Report Writing (Cover page & Introduction & Detail Steps & Detailed Description)
WONG ZI PING A25CS0164	- Prototype sketch - Prototype design

8.0 References

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[ijeter01832020.pdf](#)
- **Chanda, D.** (2025, January). *How AI Can Enhance User Experience in Sports Booking Apps?* Idea Usher.
[Use Cases And Applications Of AI In Sports - Idea Usher](#)